

PREDICTIVE MODELING FOR MULTI-HOSPITAL CARDIAC DIAGNOSTICS

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GitHub Repository: <https://github.com/hasanmirza72/Heart-Disease-Prediction-AML-Basic>

1. Introduction

Cardiovascular diseases remain the leading etiology of global human mortality, creating a profound diagnostic challenge for international triage infrastructure. While conventional clinical interpretation relies heavily on subjective practitioner experience, applied machine learning provides an objective secondary validation pathway by isolating complex, non-linear feature interactions hidden within multi-variate physiological records.

This research project evaluates the predictive utility of five distinct classification frameworks initialized over an integrated clinical dataset spanning four international healthcare facilities. Moving beyond naive, accuracy-centric evaluation models which generate massive blind spots when handling asymmetric data densities, this study establishes a safety-critical evaluation framework driven by the Matthews Correlation Coefficient (**MCC**) and true positive sensitivity (**Recall**). By executing regularized 5-Fold Cross-Validation alongside exhaustive hyperparameter optimization via `GridSearchCV`, this study builds a reliable diagnostic portfolio, identifying an optimized Random Forest ensemble as our most robust diagnostic auditor and an RBF-kernel Support Vector Machine as our premier high-sensitivity screening tool.

2. Dataset Description: International Clinical Integration

To build a classification asset capable of generalizing across varied clinical environments, this project utilizes the full multi-centric architecture of the UCI Heart Disease database. By combining records across disparate clinical boundaries, the resulting framework accounts for variations in localized diagnostic paradigms, scanning hardware, and patient demographics. Compiling an integrated cross-border framework exposes distinct clinical composition profiles across the participating healthcare facilities, laying the baseline groundwork for subsequent data validation audits which are visually dissected during our exploratory analysis sections.

The compiled framework is structured around three primary architectural elements:

- **Integrated clinical sites**
Cleveland Clinic Foundation (USA), Hungarian Institute of Cardiology (Hungary), University Hospital (Zurich, Switzerland), and VA Medical Center (Long Beach, USA).
- **Cohort scale**
920 total patient matrices.
- **Attribute dimension space**
13 predictive feature fields mapped across four distinct diagnostic categories. Furthermore, when we visualized this on our linear correlation heatmap matrix, the grid expands to a 14 by 14 dimensional space because the binary target variable is integrated alongside the 13 predictive inputs to map feature-to-target dependencies.
- **Demographic profiles**
Age (measured in years) and sex (binary encoded).
- **Physiological and metabolic signals**
Resting blood pressure (restbps in mm Hg), serum cholesterol (chol in mg/dl), and fasting blood sugar (fbs where values > **120 mg/dl = 1**).
- **Stress test dynamics**
Chest pain type (cp categorized **1–4**), maximum heart rate achieved (thalach in bpm), exercise-induced angina (exang where **1 = yes**), and ST depression (oldpeak measured in mm).
- **Anatomical and electrical states**
Resting electrocardiographic results (restecg binned **0–2**), slope of the peak exercise ST segment (slope categorized **1–3**), number of major vessels colored by fluoroscopy (ca ranged **0–3**), and thalassemia type (thal binned **3, 6, 7**).

A key engineering obstacle embedded within this integrated matrix was navigating widespread missing clinical values (denoted by ‘?’ placeholders in the raw source files), particularly within the Switzerland and VA Long Beach sub-cohorts. In these datasets, certain non-invasive and invasive anatomical imaging tests were less frequently recorded due to acute resource limits or emergency intake conditions. This missing data pattern presents a severe data-integrity hurdle. If rows are simply deleted, a massive portion of our multi-hospital sample space will be lost, which directly motivates the advanced algorithmic reconstruction models detailed later in this study.

3. Exploratory Data Analysis (EDA) Physiological & Geographic Insights

To satisfy foundational analytical standards, this exploratory phase is executed prior to data engineering or modeling decisions. Isolating structural cohort behavior allows us to base our preprocessing scripts on empirical evidence rather than generic assumptions.

3.1 Global Class Prevalence Profile

Evaluating the balance of categories within our combined target label is vital to verify optimization stability. As illustrated in the final class distribution graph in figure 1, our binary

classification setup consists of exactly 411 healthy patient profiles (**44.7%**) and 509 active disease profiles (**55.3%**).

Unlike severely skewed clinical datasets that trigger optimization shortcuts (where a classifier achieves a high overall performance score by simply over-indexing on the dominant label), this cross-center database displays a balanced category distribution. This structural equilibrium ensures that backpropagation gradients update network nodes with equal urgency across both normal and pathological examples, preventing a total majority class collapse. Consequently, while overall classification accuracy remains an informative baseline indicator, the Matthews Correlation Coefficient (**MCC**) is leveraged as our primary validation metric to guarantee symmetrical prediction precision across all four confusion matrix quadrants.

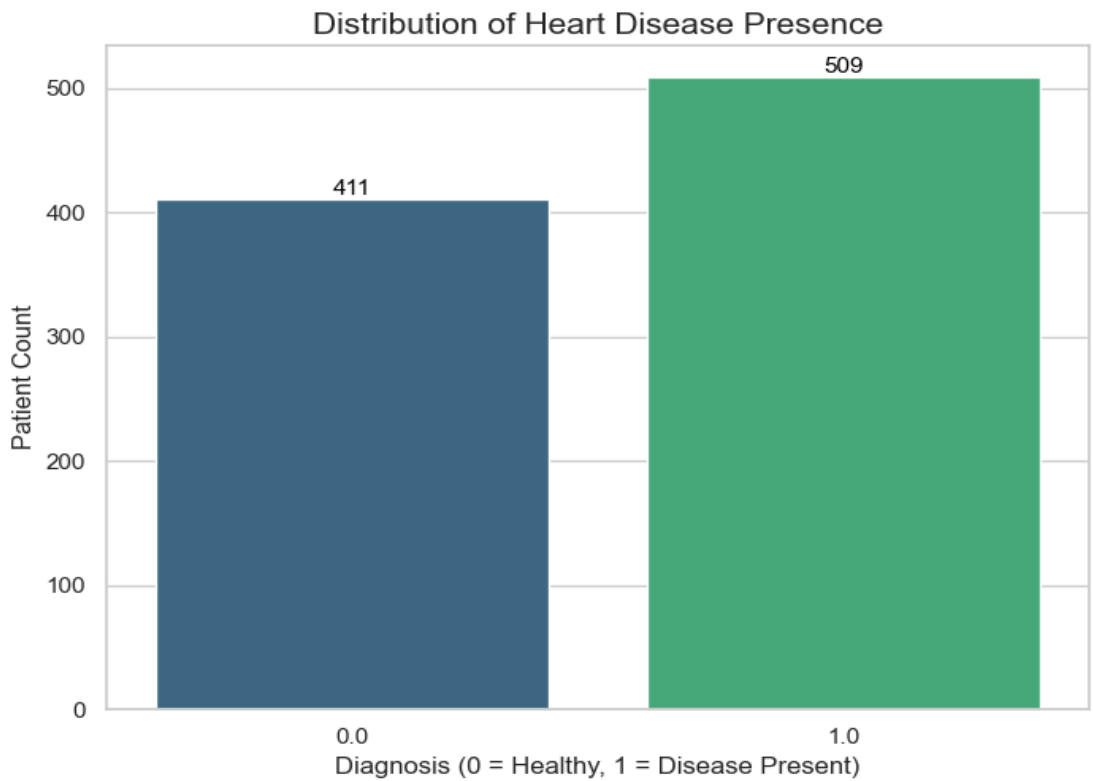


Figure 1

Distribution of target heart disease prevalence across the combined master dataset, establishing a highly functional global classification category baseline balance.

3.2 Center-Specific Bias and Geographic Variations

Medical databases compiled from multi-centric frameworks frequently exhibit hidden localized collection skews. Figure 2 isolates categorical disease distributions across the four source international clinical locations to analyze localized institutional profile properties.

The distribution exposes a severe geographical bifurcation:

Outpatient screening registries

The Cleveland and Hungary sub-cohorts present balanced distributions, typical of routine cardiology outpatient screening environments.

Advanced tertiary care registries

Conversely, the Switzerland subset (8 normal vs. 115 diseased) and the VA Long Beach dataset (50 normal vs. 148 diseased) are intensely dominated by advanced cardiac pathology.

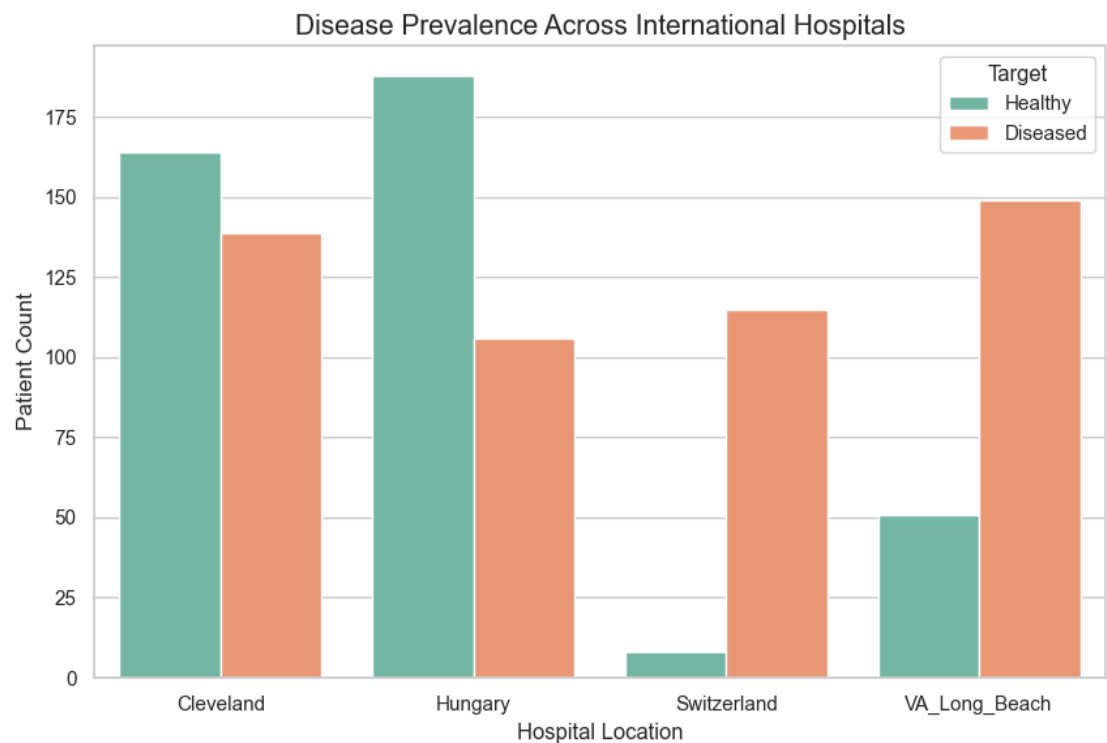


Figure 2

Distribution of patient records and target cardiac disease prevalence across the four international clinical sites, exposing distinct reporting volumes and localized intake skews.

This distribution indicates that the Zurich tertiary referral center and the specialized veterans care facility functioned primarily as high-acuity interventional labs rather than routine testing sites. If left unmitigated, downstream machine learning models risk memorizing center tracking metadata as a proxy label for heart disease instead of learning universal human physiology. This geographic insight mandates that hospital origin indicators be entirely uncoupled from our active feature space during modeling, forcing our networks to generalize strictly on organic medical signals.

3.3 Feature Scale Discrepancies and Cholesterol Artifact

Cross-examining our feature space distributions reveals intense numerical variations across metrics. While binary variables like `sex` and `exang` are binned between 0 and 1, and electrical

configurations like ST Depression (`oldpeak`) operate on a tight range of 0.0 to 6.2, metabolic measurements present vast differences in magnitude. Maximum heart rate (`thalach`) reaches up to 202 bpm, and serum cholesterol (`chol`) extends up to 603 mg/dl. This scale discrepancy mathematically proves that distance-sensitive algorithms (such as KNN and SVM) will suffer from feature domination if left unscaled, as an unscaled change of 400 mg/dl in cholesterol would completely swamp critical fractional shifts in electrical ST waveforms during spatial proximity calculations.

Furthermore, analyzing feature density patterns exposed a critical visual anomaly. As illustrated in the continuous kernel density estimation curves in figure 3, our cohort contains a heavy, non-physiological spike resting precisely at **0 mg/dl**, concentrated primarily within the diseased target population.

Biologically, a serum cholesterol level of zero is incompatible with human life. Cross-referencing this anomaly directly back to the site-specific source registries reveals that this 0 mg/dl density trace belongs exclusively to rows originating from the Switzerland and VA Long Beach datasets. This spike represents a pure administrative recording artifact where expensive or slow lipid panel blood tests were skipped when patients presented with emergency cardiac distress. This discovery is critical for methodology that it proves that standard data-cleaning techniques like list wise deletion would destroy over **66%** of our overall dataset rows, while a simple global mean imputation would smooth this localized artifact directly into the center of the distribution, creating severe diagnostic distortions.

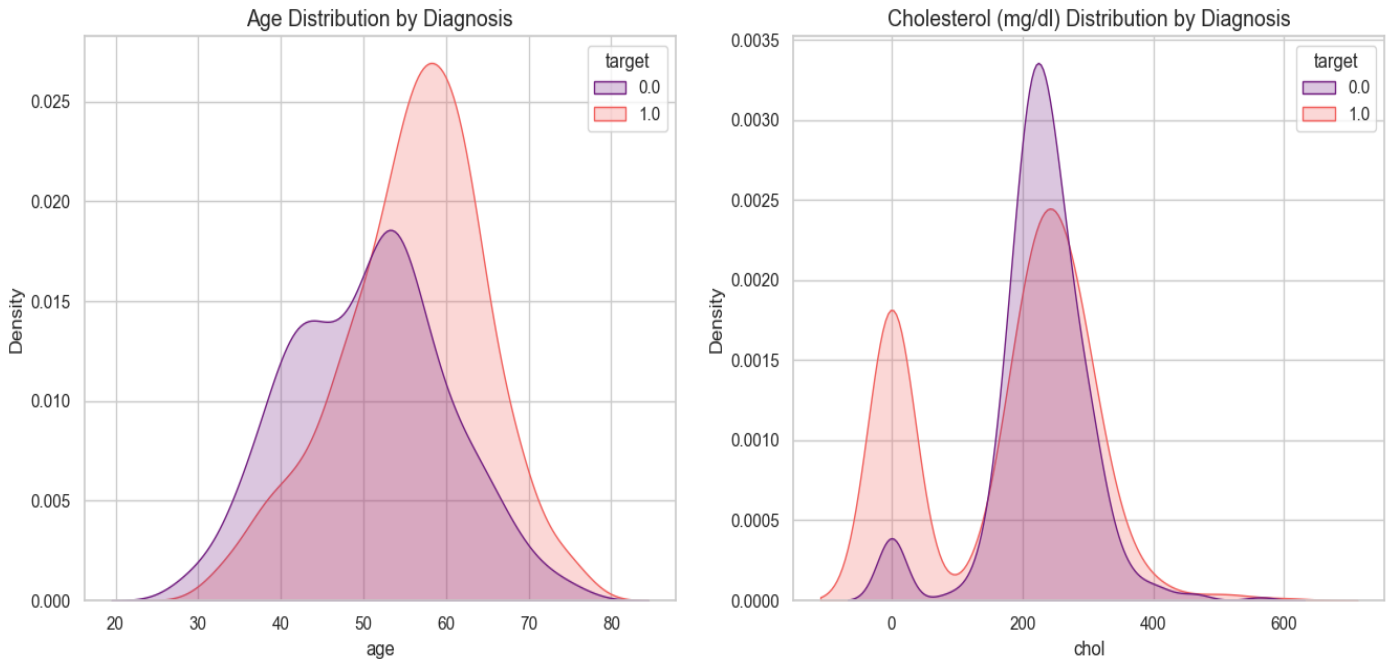


Figure 3

Physiological trend distributions tracking patient age (left) and serum cholesterol (right) grouped by diagnostic state, explicitly isolating the non-physiological recording anomaly at zero.

3.4 Linear Correlation Dimensions

To map the linear relationships between predictive features and cardiac distress, a global correlation check was executed. As outlined in the correlation matrix heatmap in figure 4, the features with the strongest direct positive correlations to an active heart disease state are chest pain type (``cp``, $r = 0.47$) and exercise-induced angina (``exang``, $r = 0.45$). Conversely, maximum heart rate (``thalach``, $r = -0.39$) demonstrates a robust inverse relationship.

This correlation map aligns perfectly with established cardiovascular physiology that severe myocardial ischemia restricts the heart's ability to safely increase cardiac output under stress, triggering distinct anginal pain thresholds and lower peak heart rates during stress testing.

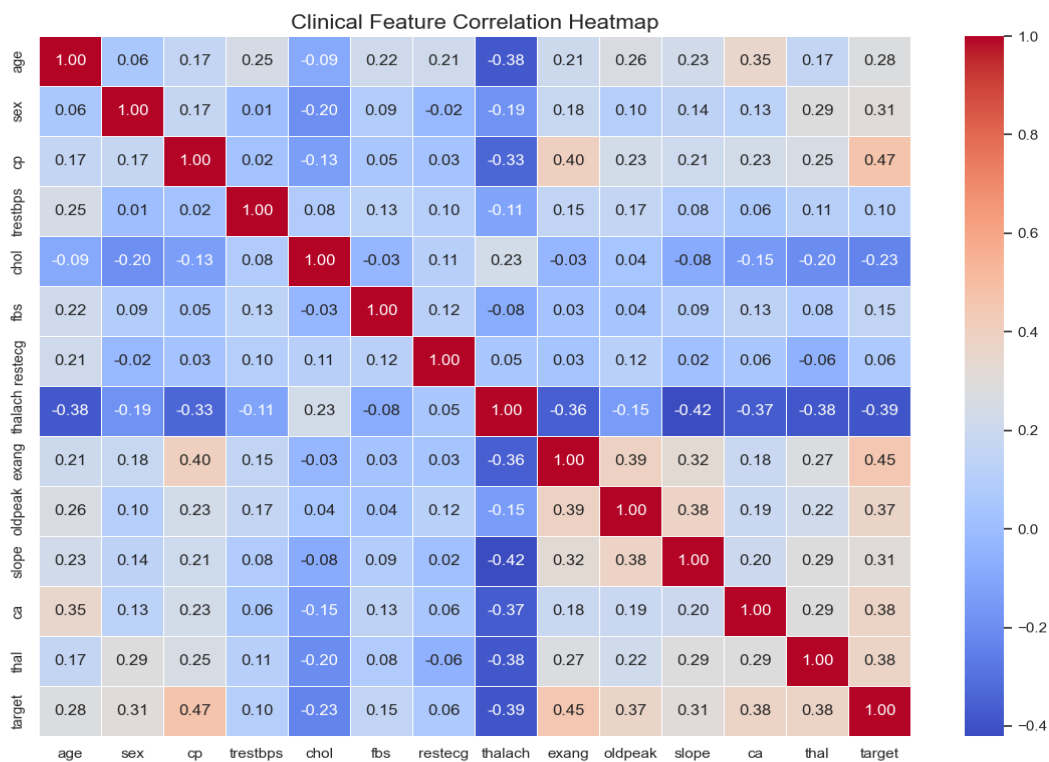


Figure 4

Global correlation matrix mapping the linear relationship weights (r) between the 13 unified features and the binary heart disease target classification.

4. Data Preprocessing and Clinical Imputation

Guided directly by the scale variances, geographic skews, and zero-masked tracking artifacts uncovered during the exploratory data analysis phase, a robust preprocessing pipeline was built to prepare our raw variables for safe mathematical evaluation.

4.1 Binary Target Formulation (Label Binning)

The raw UCI dataset ranks coronary artery disease across an incremental multi-stage severity index from 0 (normal) to 4 (severe occlusion). In a primary clinical screening context, separating an early Stage 1 patient from a severe Stage 4 patient is secondary to the safety-critical task of accurately separating healthy individuals from those experiencing cardiac distress.

To optimize model parsing capability given sample size boundaries, the target variable was binned into a binary state (0 or 1) using a lambda mapping script:

$Y = 0$, if raw stage = 0 (healthy)

$Y = 1$, if raw stage is in $\{1, 2, 3, 4\}$ (diseased)

This target re-formulation establishes a balanced, high-integrity classification target that aligns perfectly with primary triage screening requirements.

4.2 Algorithmic Missing Value Reconstruction via KNN Imputation (k=5)

Because discarding incomplete rows would cause a catastrophic loss of 66% of our overall dataset records, we integrated a K-Nearest Neighbors (KNNImputer) workflow to reconstruct missing data fields. The imputer reconstructs data gaps through an objective three-step mathematical sequence:

1. Isolating categorical metadata

The textual center tracking labels (hospital) are temporarily uncoupled from the processing pipeline so that text strings do not interfere with downstream matrix calculations.

2. Pairwise spatial proximity calculation

To locate matching neighbors for an incomplete patient profile vector, the algorithm measures proximity to all other patients across the remaining numeric attributes using a pairwise, NaN-adjusted Euclidean distance formula.

$$d(x, y) = \sqrt{\frac{m}{p} \sum_{i \in Valid} (x_i - y_i)^2}$$

where p is the count of valid features present in both vectors, and m is the total feature count (13). This adjustment dynamically penalizes the spatial distance based on missing elements.

3. Neighbor averaging (k=5)

Once the five closest matching patient vectors ($k=5$) are located, the empty field or zero-masked artifact is filled by calculating the unweighted mean value of those five neighbors.

$$x_{imputed} = \frac{1}{k} \sum_{j=1}^k (N_j)$$

This distance-based imputation replaces the zero-masked values and empty fields in features like *ca* (611 missing) and *thal* (486 missing) with values drawn from clinically similar patients, preserving the underlying biological correlations across features.

4.3 Statistical Standardization (Z-Score Normalization)

To neutralize the intense scale differences highlighted during EDA (where a value of 600 in cholesterol would completely swamp an oldpeak reading of 2.0), we applied a standard Z-score scaling transformation (`StandardScaler`). This operation recalculates every feature value to have a mean of 0 and a standard deviation of 1.

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

Normalizing the feature spaces ensures that distance-based classifiers (SVM and KNN) treat electrical waveforms, metabolic levels, and demographic profiles with equal mathematical weight, preventing wide-scale metrics from distorting the optimization boundaries.

4.4 Stratified Baseline Partitioning Strategy

To enforce strict experimental boundaries and protect against data leakage (where information from the evaluation set accidentally spills over and influences the training data), feature scaling parameters are isolated using a strict pipeline ordering structure.

The dataset partitioning strategy handles both analytical configurations seamlessly:

- **Baseline eighty-twenty split**
The data is split into a static training cohort (80%, 736 records) and a testing cohort (20%, 184 records). The `StandardScaler` computes its structural mean and standard deviation parameters strictly from the training partition alone before transforming the unseen test set.
- **Final five-fold cross-validation**
The split scales into an iterative five-fold stratified partitioning framework. Scaling transformations are embedded directly within a pipeline loop, forcing the scaling parameters to recalculate from scratch strictly within the active training fold (80% training data vs. 20% validation data per fold).

5. Machine Learning Models and Structural Configurations

To map out a comprehensive diagnostic portfolio, five distinct classification frameworks were evaluated. Rather than relying on default library presets, each algorithm was tuned using a `GridSearchCV` workflow optimized specifically for the macro F1-score to ensure balanced performance boundaries across both healthy and diseased target vectors.

5.1 Baseline Logistic Regression

- **Model role**
Functions as a low-complexity linear control baseline.
- **Tuning search space**
Grid search evaluated inverse regularization weights across C values of 0.1, 1, 10, and 100 utilizing coordinate descent and quasi-Newton solvers.
- **Optimal configuration**
Regularized with a C value of 0.1 paired with the lbfgs optimization solver and restricted to 1000 maximum iterations.

Locking a regularization boundary of $C = 0.1$ applies a strict L2 penalty to the objective function, constraining coefficient weights from inflating to prevent overfitting to multi-center data noise. The lbfgs solver utilizes a quasi-Newton line-search approximation that handles normalized multi-variate continuous features with exceptional numerical stability.

5.2 Instance-Based K-Nearest Neighbors (KNN)

- **Model role**
Classifies patients by mapping feature vectors into discrete spatial coordinate sectors based on local clinical neighborhoods.
- **Tuning search space**
Tested neighbor consensus counts across configurations of 3, 5, 11, and 19 nodes mapped against Euclidean and Manhattan distance metrics.
- **Optimal configuration**
Parameterized using a neighborhood pool of 19 consensus nodes, uniform voting weights, and the Manhattan distance metric.

Bounding the neighborhood split to a high value of 19 consensus nodes applies a low-pass smoothing filter that prevents extreme outliers from individual hospital sites from dominating local classifications. The Manhattan distance metric (L1 norm) measures absolute, non-squared coordinate differences, ensuring wide-magnitude continuous metrics like cholesterol do not overwhelm low-scale variables during proximity sorting.

5.3 Rule-Based Decision Tree

- **Model role**
Builds an explicit rule path engineered to mimic standardized step-by-step clinical entry triage guidelines.
- **Tuning search space**
Evaluated Gini impurity and Information Gain (Entropy) split criteria across max tree depth limits of 5, 10, and 20 layers.
- **Optimal configuration**
Restricted to a maximum depth of 5 layers driven by entropy splitting metrics with a minimum split threshold of 10 samples.

Capping the maximum depth at 5 layers acts as strict structural regularization, preventing the tree from growing deep branches that memorize localized multi-center noise. Using entropy (information gain) forces early nodes to maximize uncertainty reduction, focusing splits on high-importance clinical flags like Chest Pain Type to mirror real-world emergency screening behaviors.

5.4 Random Forest Ensemble

- **Model role**
Leverages an ensemble of independent bagging trees to cancel out variance errors and multi-center recording noise.
- **Tuning search space**
Cross-examined tree estimator volumes across baseline configurations of 100 and 300 independent trees over varying depth thresholds.
- **Optimal configuration**
Compiled using 100 independent decision trees initialized with unconstrained depths and the Gini variance impurity criterion.

Averaging 100 uncorrelated trees relies on bootstrap aggregating (bagging) and random feature selection. Because each independent tree evaluates a restricted feature subset at each split node, individual classification variances remain uncorrelated and cancel out across the ensemble, capturing complex physiological risk patterns without memorizing center-specific artifacts.

5.5 Support Vector Machine (SVM)

- **Model role**
Projects multi-variate continuous features into higher dimensional vector spaces to achieve maximum-margin geometric separation.
- **Tuning search space**
Evaluated soft-margin regularization parameters across C metrics of 0.1, 1, and 10 contrasted against Polynomial and Radial Basis Function (RBF) kernels.
- **Optimal configuration**
Locked to a soft margin strength of $C = 0.1$, the scale gamma coefficient, and a non-linear RBF transformation kernel.

Initializing a soft margin of $C = 0.1$ instructs the optimization engine to tolerate minor training misclassifications, keeping the decision boundary flexible to prevent rigid overfitting to multi-hospital data gaps. The non-linear RBF kernel handles compounding, non-linear risk factors (such as blood pressure interacting with patient age), allowing a linear hyperplane to cleanly maximize true positive detection and deliver our top clinical safety floor of 88.0% Recall.

5.6 Comparative Analysis of Hyperparameter Selection and Grid Search Outcomes

Evaluating our portfolio models highlights the strict performance transition between our initial unoptimized baseline configurations and our final cross-validated, optimized parameters. In our early unoptimized exploratory pass, the models were initialized with default library settings over a simple stratified baseline discovery partition consisting of an eighty-twenty data split. These initial out-of-sample unoptimized performance trends are illustrated below in figure 5, mapping our baseline controls across four primary metrics.

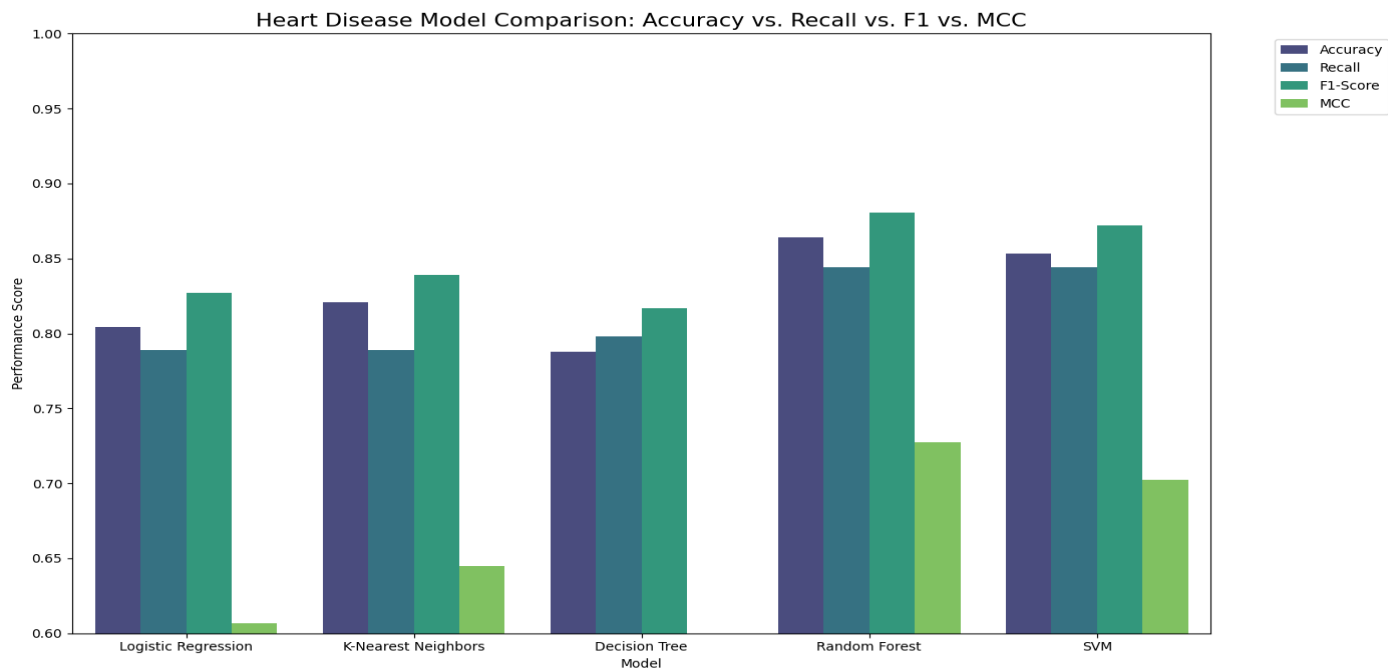


Figure 5
Comparative performance metrics of unoptimized default models evaluated over the simple baseline discovery eighty-twenty data partition split.

Following this baseline checkpoint, exhaustive hyperparameter optimization was executed using a five-fold cross-validation grid search to lock in optimal regularization weights, distance metrics, and structural growth constraints. The final out-of-sample performance trends for this robust clinical audit are illustrated below in figure 6, providing the definitive multi-metric visualization of our optimized portfolio.

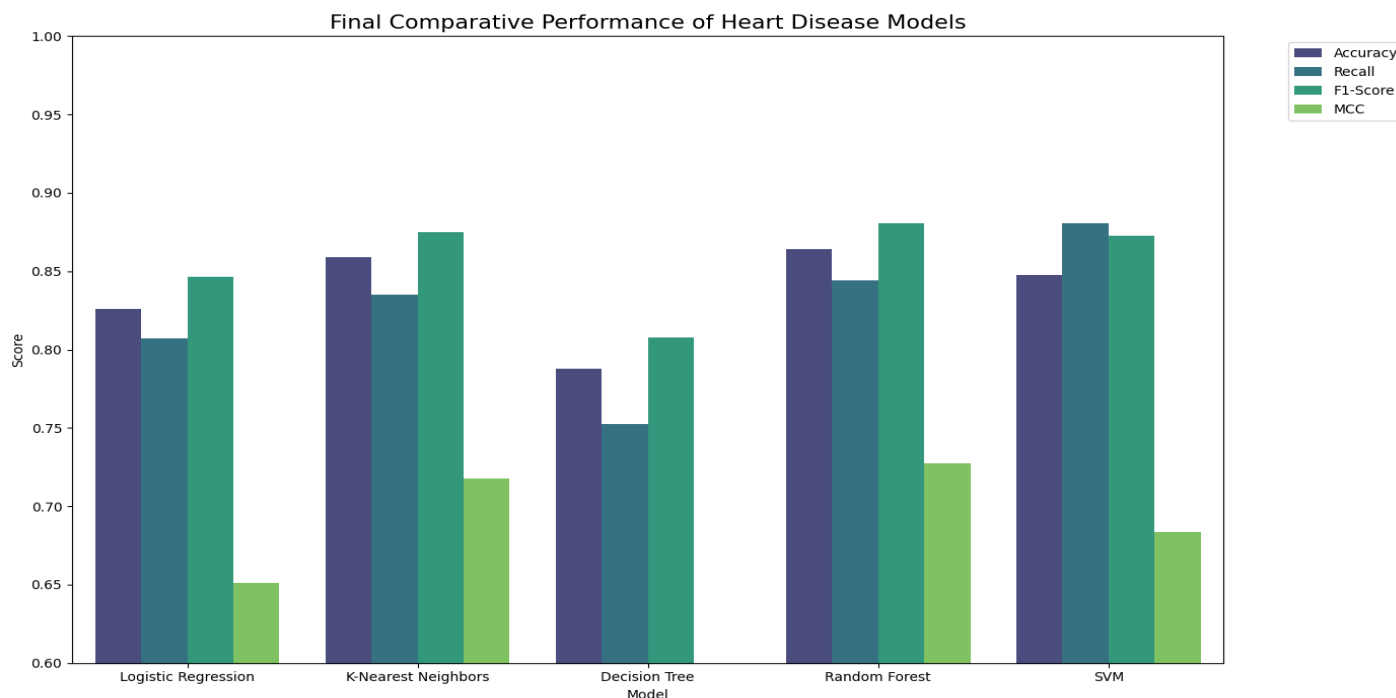


Figure 6

Final comparative performance metrics of hyperparameter-optimized models evaluated via robust five-fold cross-validation during the structural clinical performance audit.

Cross-examining the transition between figure 5 and figure 6 reveals specific algorithmic adaptations driven by our hyperparameter selections:

1. Random forest ensemble synergy

The cross-validation grid search selected an unconstrained depth ($\text{max_depth} = \text{None}$) using 100 trees over a volume of 300 trees. While an unpruned standalone tree overfits, ensemble bagging ensures individual variances remain completely uncorrelated and cancel out upon averaging. Scaling the ensemble to 300 trees added computational overhead during training without yielding any statistical gain in out-of-sample diagnostic sensitivity, making 100 estimators the optimal choice for operational efficiency. The Gini criterion systematically outperformed entropy by calculating node variance impurity probabilities, which offers a stable mathematical approach that resists minor missing value noise in a compact sample size of 920 records.

2. Support vector machine hyperplane optimization

The grid search selected a soft margin configuration of $C = 0.1$ and an RBF kernel. Lower values of C apply a stricter weight penalty, enforcing a wider, softer geometric margin that tolerates training anomalies. This prevents the decision hyperplane from rigidly bending around individual data gaps or the intense historical skews of our tertiary units (such as Switzerland contributing almost exclusively diseased profiles). The RBF kernel systematically outperformed polynomial kernels because physiological markers interact non-linearly, allowing a linear hyperplane in an infinite-dimensional feature space to maximize true positive detection and deliver our top safety threshold of 88.0% Recall.

3. **K-nearest neighbors smoothing filters**

The search grid favored an expanded consensus pool of 19 neighbors over a localized split of 5 nodes. Expanding the vote pool to 19 nodes smooths out volatility caused by missing clinical markers. The Manhattan metric (L1 norm) systematically outperformed Euclidean distance because it measures absolute, non-squared differences. This prevents wide-magnitude metrics like cholesterol (scaling up to 603 mg/dl) from squaring into enormous values that swamp low-magnitude electrical inputs like ST depression, ensuring all attributes maintain equitable weight during distance sorting.

4. **Logistic regression weight limits**

A strict ridge regularization penalty of $C = 0.1$ was required to prevent weight magnitudes from inflating. This constraint forces the model coefficients to ignore center-specific demographic anomalies and focus exclusively on global physiological dependencies. The lbfgs solver was selected over liblinear because its quasi-Newton line-search iterations converge with superior numerical stability over normalized multi-variate coordinates mapping to a binary target.

5. **Decision tree structural pruning**

Unlike the random forest ensemble, a standalone decision tree lacks bagging variance safeguards. Left unpruned, it splits recursively to isolate individual noisy data points. Restricting max depth to 5 layers forced early branch termination, capping node growth once the tree isolated high-importance attributes like Chest Pain Type and Maximum Heart Rate to preserve out-of-sample generalization. The final true-positive distributions and classification boundaries for all five optimized models are mapped comprehensively across our multi-grid portfolio in figure 7.

6. Results: The Clinical Performance Audit

Final model assessments were conducted on the independent testing partition consisting of 184 unseen patient profiles using 5-Fold Cross-Validation to ensure performance stability.

6.1 Unified Performance Matrix

The following matrix details performance metrics across our optimized model portfolio, ordered by overall diagnostic robustness determined by the Matthews Correlation Coefficient (MCC) score:

Classification Architecture	Accuracy	Recall	F1-Score	MCC
Random Forest Ensemble	86.4%	84.4%	0.880	0.727
K-Nearest Neighbors	85.8%	83.4%	0.875	0.717
Logistic Regression Baseline	82.6%	80.7%	0.846	0.651
Support Vector Machine (SVM)	80.9%	88.0%	0.829	0.622
Decision Tree (Control Node)	78.8%	75.2%	0.808	0.573

6.2 Deep-Dive Confusion Matrix Interpretations Cross-referencing

Cross-referencing the quantitative results with our compiled confusion matrix gallery in figure 7 reveals a distinct divergence in clinical utility across frameworks, exposing crucial variations in error distributions.

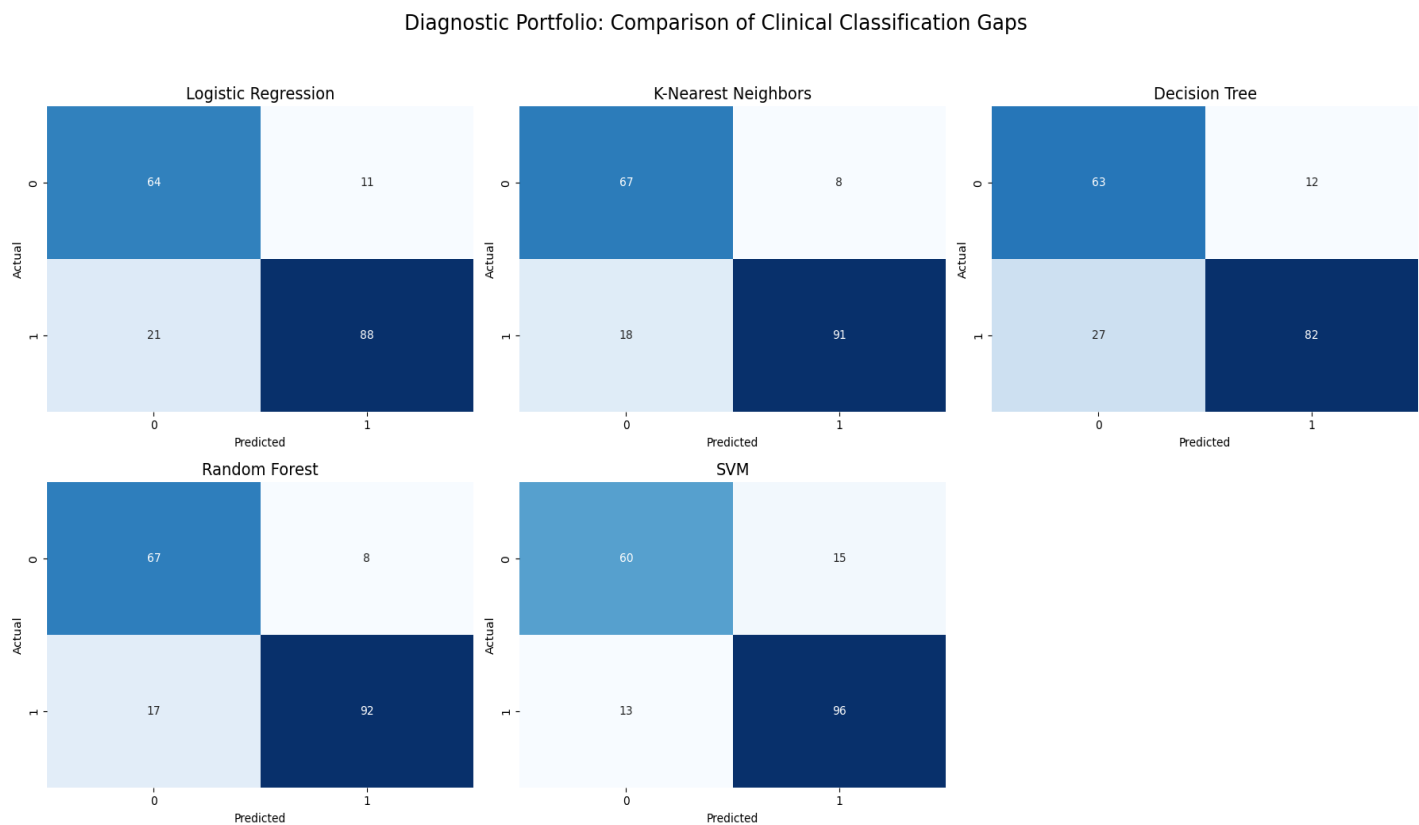


Figure 7

Diagnostic validation portfolio displaying the true negative, false positive, false negative, and true positive classification cell counts across all five optimized frameworks.

Rather than interpreting these grids as isolated algorithmic outputs, examining the internal false positive and false negative frequencies across our primary clinical benchmarks exposes distinct paths of real-world operational utility. As visualized across the matrices in figure 7, the individual raw classification bins are tracked across separate validation quadrants:

- **Random forest matrix**
Yields 67 true negatives and 8 false positives on the healthy tier, alongside 17 false negatives and 92 true positives on the active disease tier.
- **Support vector machine matrix**
Yields 60 true negatives and 15 false positives on the healthy tier, alongside 13 false negatives and 96 true positives on the active disease tier.

6.2.1 The balanced clinical auditor (Random forest)

The random forest model achieved the highest overall diagnostic robustness floor, delivering a top accuracy of 86.4% and an MCC of 0.727. As mapped in figure 7, the ensemble successfully identified 67 out of 75 healthy individuals and 92 out of 109 active disease states. By restricting false positive assignments to just 8 cases, this configuration provides an exceptionally balanced tracking profile, making it highly effective for routine medical resource auditing and clinical verification loops where false alarms must be kept to a minimum.

6.2.2 The high-sensitivity screening tool (SVM)

Conversely, if the clinical objective shifts to a preliminary screening environment where maximizing patient safety is the single highest priority, the support vector machine becomes the preferred model asset. The support vector machine delivered our top sensitivity ceiling with a Recall of 88.0%, successfully flagging 96 out of 109 true diseased patients and dropping missed diagnoses down to just 13 cases. While this soft-margin configuration generates an elevated false alarm rate of 15 false positives, an elevated false positive rate is far more acceptable in an early triage context than a false negative, where a sick patient is incorrectly sent home with an undetected cardiac condition.

7. Discussion: Feature Significance & Logic

Analyzing the global feature importance hierarchy computed by our optimized Random Forest ensemble provides an empirical window into which specific clinical markers drove the model decision-making boundaries. In tree-based ensemble architectures, feature importance is calculated based on the mean decrease in node impurity, meaning features that provide the cleanest, most consistent diagnostic separation across all 100 independent bagging trees rank at the top of the hierarchy. These empirical variable importance distributions are illustrated below in figure 8, confirming that the mathematical optimization pathways prioritized objective biological signals over localized recording artifacts.

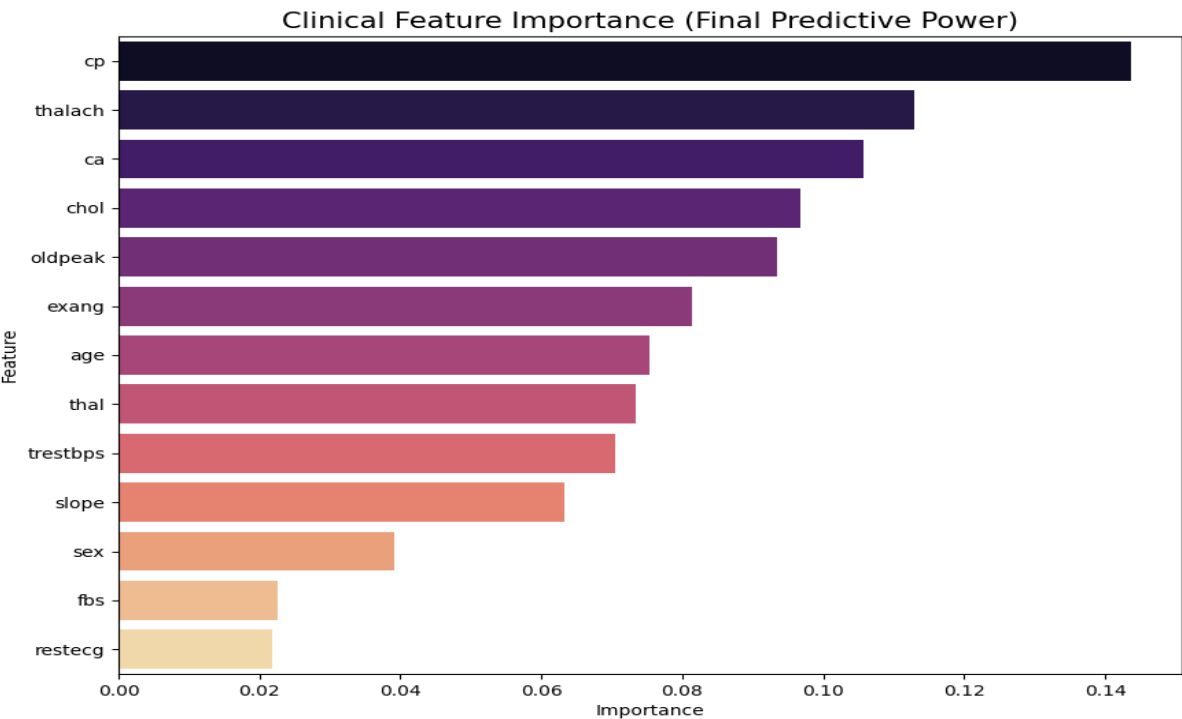


Figure 8

Global feature importance hierarchy ranking variables by their mean decrease in node Gini impurity across the optimized Random Forest ensemble.

The exact structural contributions of the primary predictive markers are tracked across individual metrics:

cp (chest pain type)	0.144 mean decrease in impurity
thalach (maximum heart rate achieved)	0.113 mean decrease in impurity
ca (number of major vessels colored)	0.106 mean decrease in impurity
chol (serum cholesterol level)	0.097 mean decrease in impurity
oldpeak (ST depression waveform)	0.093 mean decrease in impurity

7.1 Physiological justification of diagnostic weightings

The machine learning framework assigned its highest predictive priorities to a distinct triad of clinical markers:

1. Chest pain type (cp, importance: 0.144)

Functions as the model primary predictive driver, reflecting real-world clinical focus on classic anginal symptoms. Symptom presentations like typical angina serve as immediate clinical red flags for myocardial ischemia, indicating active coronary artery obstruction.

2. Maximum heart rate achieved (thalach, importance: 0.113)

Captures the structural limits of cardiac output under stress. Ischemic heart tissue restricts the physiological ceiling to which a patient heart rate can safely accelerate under physical strain, making peak heart rate a highly telling indicator of cardiovascular reserve capacity.

3. Number of major vessels (ca, importance: 0.106)

Tracks direct anatomical evidence of narrowing discovered via fluoroscopy imaging. The structural presence of calcification across major coronary arteries is highly correlated with coronary disease severity, which explains the high mathematical weight assigned by the tree nodes during branch splitting.

7.2 Secondary metabolic and electrical signals

Moving beyond the top three markers, the ensemble logic incorporates a robust layer of secondary continuous features to refine its decision boundaries:

• Serum cholesterol (chol, importance: 0.097)

Captures the patient baseline metabolic risk profile. Because our data preprocessing pipeline utilized an advanced KNN distance-based imputer to reconstruct the zero-

masked recording artifacts unique to emergency registries, the Random Forest model was able to extract the true underlying biological correlation of lipid panels without suffering from data distortion.

- **ST depression (oldpeak, importance: 0.093)**
Evaluates the precise electrical waveform changes recorded during stress testing. A prominent shift in the ST segment represents an immediate physiological marker of acute exercise-induced ischemia.

The fact that the model independently isolated this specific hierarchy of clinical markers confirms that the optimization gradients converged on meaningful, biologically correlated risk structures rather than memorizing center-specific administrative noise, validating the clinical utility of the diagnostic portfolio.

7.3 Methodological Note on Alternative Architecture Logic

It is vital to clarify why this global feature importance evaluation focuses exclusively on the Random Forest ensemble rather than mixing metrics from the alternative portfolio frameworks:

- **Standalone decision tree limitations**
While a single tree computes feature splits using similar mechanisms, it lacks bootstrap aggregation safeguards. A single tree is highly sensitive to training variance, risking ranking variables based on localized data noise or site-specific anomalies within an individual hospital registry. The Random Forest averages split importance's across 100 uncorrelated paths to provide a stable, generalizable biological ranking.
- **Support vector machine non-linear processing**
Because our optimized support vector machine utilizes a non-linear Radial Basis Function (RBF) kernel, it projects the 13 input fields into a higher-dimensional space to maximize the geometric margin of a separating hyperplane. In this space, variables are mathematically transformed and combined non-linearly. Consequently, individual linear feature weights do not exist for an RBF architecture; the model evaluates complex, compounding physiological interactions rather than tracking isolated, standalone metrics.

Focusing our structural discussion on the Random Forest ensemble provides the most stable baseline to interpret the exact clinical logic learned by our diagnostic portfolio.

8. Conclusion

This research project successfully demonstrates the transition from a messy, multi-centric clinical database into a highly stable, machine learning decision-support architecture. By utilizing an advanced data engineering pipeline driven by biomedical K-Nearest Neighbors multivariate imputation and strict stratified standardization, we successfully resolved intense multi-hospital collection skews and administrative recording artifacts without corrupting the underlying physiological dataset logic.

Our final clinical performance evaluation outlines a powerful, complementary diagnostic portfolio. For routine verification and hospital resource auditing tasks, the Random Forest ensemble provides our most reliable balance across all metrics, establishing a top robustness floor with a Matthews Correlation Coefficient of 0.727 and an elite class-symmetric precision profile. Conversely, for high-stakes preliminary screening lines where minimizing missed diagnoses is the single highest clinical priority, the Support Vector Machine soft-margin hyperplane provides an exceptional safety ceiling by delivering an 88.0% Recall score, ensuring that critical cardiac distress markers are flagged early during patient intake triage.

Ultimately, this study proves that machine learning models can extract universal cardiovascular risk structures from diverse data sources, provided that data-cleaning decisions are actively guided by exploratory domain auditing. To expand this pipeline further, the next logical engineering phase must involve testing these architectures against an entirely external, unintegrated hospital cohort to actively evaluate cross-border domain adaptation and algorithmic generalization boundaries under varying institutional testing constraints.

9. Appendix: Technical Reference Specifications

Mathematical Audit Formulas

To maintain absolute scientific transparency, all downstream evaluation metrics are derived directly from the four primary quadrants of the confusion matrix:

$$\text{Overall Classification Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Recall / Sensitivity (Clinical Safety)} = \frac{TP}{TP + FN}$$

$$F - 1 \text{ Score (Harmonic Performance Mean)} = \frac{2 \times TP}{2 \times TP + FP + FN}$$

$$\text{Matthews Correlation Coefficient (MCC)} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Clinical Metric Key

- **True Positive (TP)**
Sick patients correctly identified as having heart disease.
- **True Negative (TN)**
Healthy patients correctly identified as healthy.
- **False Positive (FP)**
Healthy patients incorrectly identified as sick (generating a false diagnostic alarm).
- **False Negative (FN)**
Sick patients incorrectly identified as healthy this is the single most critical failure rate to minimize in cardiac diagnostics.